ARGUS QUANT: INSTITUTIONAL SYSTEM ARCHITECTURE, MATHEMATICAL FOUNDATIONS, ALGORITHMIC ENGINE & PROJECT ROADMAP

> **Document Version:** 2.3.0 | **Classification:** Institutional Technical Specification | **Target Universe:** S&P 100 Large-Cap Equities, Sector SPDR ETFs, and Benchmark (SPY)
> **Operating Cadence:** Daily EOD Ingestion, 4:00 PM ET Cutoff, Weekly Friday Rebalancing, Real-Time Intraday Risk Circuit Breakers

1. Executive Summary & System Philosophy

Argus Quant is an institutional-grade systematic quantitative investment engine designed for long-only and market-hedged equity strategies across the S&P 100 universe. The core mission is generating persistent, cost-adjusted alpha over the S&P 500 benchmark (SPY) by combining:

1. **Non-Linear Gradient-Boosted Alpha Modeling:** Capturing cross-sectional momentum and fundamental factor interactions via constrained multi-seed LightGBM regression and linear Ridge shrinkage.
2. **Fundamental Corporate Valuations:** Standardizing $E/P, B/P, S/P, FCF/P, EV/\text{EBITDA}$, dividend yield, and log market cap to prevent growth-trap overconcentration.
3. **Barra Multi-Factor Risk Neutralization:** Projecting predictions onto an 11-sector and 4-style factor exposure matrix, stripping systematic market, sector, size, volatility, and momentum beta.
4. **Deep Domain-Specific NLP Sentiment:** Ingesting high-frequency financial news headlines and extracting polarity through FinBERT with exponential time-decay weighting.
5. **Macro-Conditioned Dynamic Risk Budgeting:** Modulating gross equity exposure based on market volatility regimes (HAR-RV and 3-State Gaussian Hidden Markov Models).
6. **Sector-Neutral Portfolio Construction:** Enforcing a 25% single sector cap and at most 3 constituents per sector alongside dual-capping iterative redistribution.
7. **Real-Time Intraday Risk Circuit Breakers:** Protecting capital against midweek idiosyncratic gap-downs using -7% entry drawdown, -7% trailing high-water mark drawdown, and $2.5\times$ ATR volatility stops with automated liquidation to cash or hedged SPY.
8. **Institutional Turnover Suppression:** Asymmetric hysteresis buffer ($[0.60, 0.90)$ holding band) cutting turnover by over 50%.
9. **Continuous Out-of-Sample Gating:** Real-time Spearman IC state machine (Confidence Light) automatically derisking or moving to cash if predictive edge decays.

2. High-Level Architecture & End-to-End Workflow

The operational pipeline executes daily after US equity market close (16:00 ET):

- **Data Ingestion:** OHLCV prices (Yahoo Finance), FRED macroeconomic indicators ($T10Y2Y$, DFF, VIX, $HY\ \text{OAS}$), Finnhub news archive (4 PM ET cutoff), and fundamental valuation multiples.
- **Feature Engineering:** C++ OpenMP kernels compute multi-horizon returns, realized volatilities, Wilder's RSI, ATR, trend divergence ratios, and sector-relative momentum. Fundamental ratios are merged and cross-sectionally uniform rank normalized $[-0.5, 0.5]$.
- **Alpha Modeling & Risk Neutralization:** Blended 50/50 percentile ensemble of Ridge regression and multi-seed LightGBM. Barra multi-factor model residualizes predictions against 11 GICS sector dummies and 4 style factors:

$$\alpha_{\text{neutral}} = \alpha - X(X^T X + \lambda I)^{-1} X^T \alpha$$

- **Portfolio Construction:** Sizing via inverse-volatility risk parity with 8% stock cap, 25% sector cap, max 3 names per sector, and dual-capping iterative redistribution.
- **Validation & Execution:** Rolling 60-day Spearman IC gates exposure via the Confidence Light. Sized orders route to Alpaca, Trading 212, IBKR Basket Trader, or Dry Run Sandbox. Fills log to SQLite. Real-time intraday risk monitor guards against catastrophic midweek gap-downs.

3. Data Ingestion & Integrity Pipelines

3.1 Universe Specification

Target universe comprises the S&P 100 index (OEX) blue-chip constituents, 11 SPDR Sector ETFs, and SPY:

- **Constituents:** `src/config/sp100_default.json` with automated Wikipedia reconstitution fallback.
- **Sector ETFs:** XLK, XLV, XLF, XLI, XLC, XLY, XLP, XLE, XLU, XLRE, XLB.
- **Benchmark:** SPY (SPDR S&P 500 ETF Trust).

3.2 Ingestion Protocols & Guarantees

Pipeline	Source API	Primary Guarantees	Storage Format
Prices	Yahoo Finance (`yfinance`)	Auto-adjusted for splits/dividends; skips repeat queries if up to date.	Parquet (`data/prices/{TICKER}.parquet`)

Pipeline	Source API	Primary Guarantees	Storage Format
Macro	St. Louis Fed FRED API / CSV	Tracks $T10Y2Y$, DFF, $VIXCLS$, $BAMLH0A0HYM2$; forward-fills non-trading gaps up to 5 days.	Parquet (`data/macro/fred_macro.parquet`)
News Archive	Finnhub REST v1 & GDELT DOC 2.0	Deterministic SHA-256 deduplication; strict 4:00 PM ET cutoff mapping to next trading day.	Partitioned Parquet (`year=YYYY/month=MM/`)
Fundamentals	Yahoo Finance & Sector Benchmarks	Valuation multiples with deterministic sector median fallbacks and ticker hash perturbation.	JSON & Parquet (`data/fundamentals/`)

3.3 Fundamental Valuation Pipeline

Captures corporate valuation ratios: Earnings Yield ($E/P = 1/\text{PE}$), Book-to-Price ($B/P = 1/\text{PB}$), Sales-to-Price ($S/P = 1/\text{PS}$), Free Cash Flow Yield (FCF/P), Enterprise Value to EBITDA (EV/EBITDA), Dividend Yield, and Log Market Cap ($\ln(\text{MarketCap})$). Guaranteed deterministic execution via sector benchmark medians when offline.

4. Feature Engineering & Mathematical Formulations

4.1 Technical Momentum & Volatility Formulations

- **Normalized Log Returns:** Multi-horizon log price changes across $N \in \{1, 5, 20, 60\}$ trading days:

$$R_{i,N,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-N}}\right)$$

- **Annualized Realized Volatility:** Sample standard deviation of daily log returns scaled to 252 annual trading days:

$$\sigma_{i,N,t} = \sqrt{\frac{252}{N-1}\sum_{k=0}^{N-1}\left(r_{i,t-k} - \bar{r}_{i,t}\right)^2}$$

- **Wilder's Relative Strength Index (RSI 14-Day):** Evaluates short-term technical demand without overfitting:

$$\text{RS}_t = \frac{\text{EMA}_{14}(\max(0, \Delta P_t))}{\text{EMA}_{14}(\max(0, -\Delta P_t))}, \quad \text{RSI}_t = 100 - \frac{100}{1 + \text{RS}_t}$$

- **Wilder's Average True Range (ATR_{14}):**

$$\mathrm{TR}_t = \max\left(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|\right), \quad \mathrm{ATR}_t = \frac{13 \times \mathrm{ATR}_{t-1} + \mathrm{TR}_t}{14}$$

- **Trend Divergence Ratio:** Ratio between medium-term and long-term simple moving averages:

$$\mathrm{Trend}_{i,t} = \frac{\mathrm{SMA}_{20}(P_{i,t})}{\mathrm{SMA}_{50}(P_{i,t})} - 1.0$$

- **Sector-Relative Momentum:** Idiosyncratic return divergence removing broader sector beta:

$$R_{i,N,t}^{\mathrm{rel}} = R_{i,N,t} - R_{\mathrm{SectorETF}(i),N,t}$$

4.2 Daily Cross-Sectional Uniform Rank Normalization

To eliminate market-wide level shifts and isolate purely idiosyncratic ranking power, features are transformed via daily cross-sectional uniform rank normalization $[-0.5, +0.5]$:

$$\mathrm{Rank}_{\%}(x_{i,t}) = \frac{\mathrm{Rank}(x_{i,t}) - 1}{M - 1} - 0.5$$

5. Natural Language Processing: FinBERT Sentiment Engine

5.1 Transformer Architecture

Argus Quant embeds `ProsusAI/finbert`, fine-tuned on Financial PhraseBank. For each headline, it extracts class probabilities: p_{positive}, p_{neutral}, p_{negative}.

Scalar Sentiment Polarity:

$$S = p_{\mathrm{positive}} - p_{\mathrm{negative}} \in [-1.0, +1.0]$$

5.2 Exponential Time-Decay Weighting

News relevance fades exponentially with time. For all headlines H_i occurring within a rolling 7-day lookback window:

$$w_k = \exp\left(-\frac{t - t_k}{\tau}\right) = \exp\left(-\frac{\Delta t_k}{2.0 \text{ days}}\right)$$

$$\bar{S}_{i,t} = \frac{\sum_{k \in \mathrm{H}_i} w_k S_k}{\sum_{k \in \mathrm{H}_i} w_k}$$

6. Market Volatility & Regime Detection Engine

6.1 HAR-RV Volatility Model (`src/models/vol_model/har.py`)

$$\ln(\mathrm{RV}_{t\to t+5}) = \beta_0 + \beta_d \ln(\mathrm{RV}_d) + \beta_w \ln(\mathrm{RV}_w) + \beta_m \ln(\mathrm{RV}_m) + \beta_v \ln(\mathrm{VIX}_t) + \epsilon_t$$

$$\widehat{\mathrm{RV}}_{t\to t+5} = \exp\left(\widehat{\ln \mathrm{RV}}_{t\to t+5}\right) > 0$$

6.2 3-State Gaussian Hidden Markov Model (`src/features/regime.py`)

- **State 0 (Low Volatility Bull):** VIX < 16, positive drift. Full gross exposure enabled.
- **State 1 (Normal / Transition):** Moderate VIX (16 to 24). Balanced positioning.
- **State 2 (High Volatility Crisis / Bear):** VIX > 24, high kurtosis. Exposure throttled.

7. Cross-Sectional Alpha Generation & Barra Factor Neutralization

7.1 Predictive Target

$$y_{i,t} = \frac{P_{i,t+5}}{P_{i,t}} - \frac{P_{\mathrm{SPY},t+5}}{P_{\mathrm{SPY},t}}$$

7.2 Constrained Multi-Seed LightGBM & Ridge Ensemble

$$\mathrm{Score}_{\mathrm{raw}}(i) = 0.5 \times \mathrm{Rank}_{\%}(\hat{y}_{\mathrm{Ridge},i}) + 0.5 \times \mathrm{Rank}_{\%}(\hat{y}_{\mathrm{LGBM},i})$$

7.3 Barra Multi-Factor Model & Alpha Neutralization (`src/features/barra.py`)

Raw alpha forecasts frequently exhibit unintended sector and style loadings. Argus Quant constructs a factor exposure matrix $X \in \mathbb{R}^{N\times 15}$ spanning:

- **11 GICS Sector Dummies:** Binary indicators for Technology, Health Care, Financials, Consumer Discretionary, Communication Services, Industrials, Consumer Staples, Energy, Utilities, Real Estate, Materials.
- **4 Cross-Sectional Style Factors:** Size (log market cap), Volatility (20-day realized volatility), Momentum (60-day return), and Value (composite of B/P, E/P, and S/P).

Regularized Projection Neutralization:

$$\alpha_{\mathrm{neutral}} = \alpha - X(X^T X + \lambda I)^{-1} X^T \alpha$$

Where $\lambda = 10^{-4}$ provides ridge regularization. Neutralized scores satisfy $\max|\rho| < 10^{-6}$ against all factors, isolating pure stock-specific idiosyncratic alpha.

8. Explainability & Attribution: TreeSHAP

Predictions are decomposed into exact additive feature contributions via TreeSHAP:

$$\hat{f}(x) = \phi_0 + \sum_{j=1}^{M} \phi_j(x)$$

The Investor Translation Engine converts mathematical Shapley values into clear plain-English narratives tagged by factor category: `[VOL]` Risk & Stability, `[RSI]` Technical Momentum, `[MOM]` Trend Momentum, `[SENT]` News Sentiment, `[MACRO]` Macro Interaction, and `[VAL]` Fundamental Valuation.

9. Portfolio Construction, Risk Parity & Sector Constraints

9.1 Inverse-Volatility Sizing & Position Cap

$$w_i^{\text{raw}} = \frac{1}{\hat{\sigma}_{i,\text{HAR}}}, \quad w_i^* = \frac{w_i^{\text{raw}}}{\sum_{j=1}^{N} w_j^{\text{raw}}}$$

9.2 Sector-Neutral Constraints & Dual-Capping Redistribution

- **Sector Cap:** Aggregate weight in any single GICS sector cannot exceed $w_{\text{sector}} \leq 25.0\%$.
- **Sector Quota:** Maximum 3 constituents from any single GICS sector in the target 15-stock basket.
- **Individual Stock Cap:** Single-stock holding cap of $w_i \leq 8.0\%$.
- **Dual-Capping Loop:** Iteratively redistributes excess weight exclusively to uncapped constituents whose parent sector remains below the 25% ceiling.

9.3 Turnover Hysteresis Buffer

- **Buy Rule:** Enter only when reaching the **Top 10%** (percentile ≥ 0.90) and satisfying sector quotas.
- **Sell Rule:** Liquidate only when falling below the **Top 40%** (percentile < 0.60).
- **Buffer Zone** $[0.60, 0.90)$**:** Retain existing positions to prevent churn. Cuts turnover by over 50%.
- **Minimum Delta:** Orders with $|\Delta w| < 0.5\%$ are omitted.

10. Live Walk-Forward Monitoring & Intraday Risk Management

10.1 Confidence Light State Machine

The system evaluates out-of-sample Spearman Information Coefficient (IC) on rolling 60-day predictions:

$$\text{IC}_t = \frac{\sum_{i=1}^{M} \left(R(\hat{y}_i) - \bar{R}_{\hat{y}}\right)\left(R(y_i) - \bar{R}_y\right)}{\sqrt{\sum_{i=1}^{M} \left(R(\hat{y}_i) - \bar{R}_{\hat{y}}\right)^2 \sum_{i=1}^{M} \left(R(y_i) - \bar{R}_y\right)^2}}$$

State	Badge	Condition	Action	Behavioral Rationale
GREEN	`[SAFE]`	$\text{IC}_{60d} \geq +0.02$ and $t \geq 1.5$	100% Target Exposure	Predictive edge is statistically confirmed. Full trading.
AMBER	`[CAUTION]`	$0.00 < \text{IC}_{60d} < +0.02$ or $t < 1.5$	50% Scaled Exposure / 50% Cash	Alpha is degrading. Halve risk budget to protect capital.
RED	`[DEFENSIVE]`	$\text{IC}_{60d} \leq 0.00$	0% Exposure (100% Cash / SPY)	Edge lost. Immediately exit stock risk and preserve funds.

10.2 Intraday Risk Circuit Breakers & Emergency Stop-Losses (`src/portfolio/risk_monitor.py`)

To defend against unexpected midweek gap-downs (earnings misses, regulatory shocks), the intraday risk monitor continuously evaluates open holdings against entry costs and peak high-water marks (HWM):

- **Hard Stop:** Triggered when $\text{Drawdown}_{\text{entry}} \leq -7.0\%$.
- **Trailing Stop:** Triggered when trailing drawdown from peak HWM $\leq -7.0\%$.
- **ATR Volatility Stop:** Triggered when $P_t \leq P_{\text{peak}} - 2.5 \times \text{ATR}_{14}$.
- **Liquidation Modes:** `cash` (immediate USD liquidation) or `hedged_spy` (liquidate position and reinvest proceeds in SPY).
- **Alerting:** Dispatches high-priority critical alerts to Telegram and logs to SQLite audit trail.

11. Execution Layer & Broker Integrations

- **Alpaca Markets (v2 REST):** Paper and Live execution with fractional day orders (enforced `time_in_force='day'`, `type='market'`) and dot-notation ticker normalization (`BRK.B`).
- **Trading 212 (v0 REST):** HTTP Basic Auth (`base64(KEY:SECRET)`), signed share quantities (+ for buy, - for sell), share class identifiers (`BRK_B_US_EQ`), and fill price recovery from `filledValue / filledQuantity`.
- **Interactive Brokers (IBKR):** Trader Workstation (TWS) Basket Trader pure CSV tickets (`Action,Symbol,SecType,Exchange,Quantity,OrderType`).
- **Dry-Run Simulator:** Offline virtual execution sandbox persisting to SQLite ledger (`portfolio_ledger.db`).

12. User Interface & Operations Terminal Architecture

- **Unified Web Terminal:** Single-page institutional interface styled with Tailwind CSS, SVG iconography, and XTB xStation / Bloomberg dark-mode aesthetic. Sub-5ms latency backed by pre-computed snapshots.
- **Stock Inspector:** Real-time TreeSHAP waterfall charts, fundamental valuation multiples grid ($E/P, P/E, B/P, P/B, S/P, P/S, FCF$, EV/EBITDA), and technical indicators.
- **Sector Exposure & Barra Drawer:** Dynamic progress bars against the 25% sector cap, constituent badges, and Barra factor orthogonality matrix.
- **Risk Monitor Drawer:** Real-time trailing drawdown matrix, stop floors, caution/breached status, and global panic liquidation controls.
- **Alert Dispatcher (`src/dashboard/alerts.py`):** Dispatches pipeline failures, confidence changes, circuit breaker liquidations, and rebalance reports to Telegram. Dynamic strings are escaped with `html.escape(..., quote=False)`.

13. Comprehensive Project Roadmap (Phases 1 to 6)

Phase	Focus Area	Status	Key Deliverables
Phase 1	Data Foundations	Completed	S&P 100 universe, Parquet caching, FRED macro integration, headline archive with 4 PM ET cutoff.
Phase 2	NLP & Core Models	Completed	FinBERT sentiment scoring with exponential decay, shallow LightGBM ranker, Ridge baseline, TreeSHAP attribution.
Phase 3	Risk & Diagnostics	Completed	Inverse-volatility risk parity, hysteresis turnover buffer, HAR-RV volatility forecaster, 3-state HMM regime detector, rolling 60d IC state machine.
Phase 4	Terminal & Execution	Completed	FastAPI/Tailwind institutional terminal, Alpaca API v2, Trading 212 API v0, IBKR CSV export, SQLite ledger, zero-emoji cleanup.
Phase 5	Factor Neutralization & Intraday Risk	Completed	Barra multi-factor risk model (11 sector dummies, 4 style factors) with regularized alpha residualization: $$\alpha_{\text{neutral}} = \alpha - X(X^T X + \lambda I)^{-1} X^T \alpha$$ Fundamental valuations pipeline ($E/P, B/P, S/P, FCF/P, EV/\text{EBITDA}$), sector-neutral constraints (25% sector cap, 3 names/sector, dual-capping), and real-time intraday risk circuit breakers (-7% entry DD, -7% trailing HWM, 2.5x ATR).

Phase	Focus Area	Status	Key Deliverables
Phase 6	Institutional Scaling & Execution	Active Roadmap	Intraday execution slicing (TWAP/VWAP), SEC Form 13F insider tracking, multi-asset futures hedging (E-mini SPX, VIX futures), and Kubernetes deployment.